

Text functions in Excel are used to manipulate, format, and extract specific parts of text strings. These are particularly helpful for cleaning data or reformatting information like names, addresses, and codes.

1. Extraction Functions

These functions allow you to pull a specific number of characters from a text string based on their position.

- **LEFT():** Extracts a specific number of characters starting from the beginning (left side) of a text string.
 - *Example:* `=LEFT("Excel", 2)` returns "Ex".
- **RIGHT():** Extracts characters starting from the end (right side) of a text string.
 - *Example:* `=RIGHT("Excel", 2)` returns "el".
- **MID():** Extracts characters from the middle of a text string, starting at any position you specify.
 - *Example:* `=MID("Engineering", 3, 4)` returns "gine".

2. Case Formatting Functions

These functions standardize the capitalization of your data, which is useful for fixing inconsistent data entry.

- **UPPER():** Converts all letters in a text string to uppercase.
- **LOWER():** Converts all letters in a text string to lowercase.
- **PROPER():** Capitalizes the first letter of each word and converts all other letters to lowercase. This is ideal for names.

3. Joining and Cleaning Functions

These tools help you combine multiple cells or remove unnecessary elements from your data.

- **CONCATENATE() or CONCAT():** Joins two or more text strings into one.
 - *Note:* You can also use the ampersand (&) symbol to join text (e.g., `=A1 & " " & B1`).
- **TRIM():** Removes all extra spaces from a text string except for single spaces between words. This is essential for fixing data that has leading or trailing spaces that might break **VLOOKUP** or **Sorting**.
- **LEN():** Returns the number of characters in a text string, including spaces. This is often used with **Data Validation** to ensure entries like IDs meet a specific length.

4. Search and Replace Functions

- **FIND() / SEARCH():** Locates the position of one text string within another.
- **SUBSTITUTE():** Replaces existing text with new text in a string.

Practical Scenario

If you have a column of "Full Names" and need to generate email addresses, you could use a combination of these:

1. **LOWER** to make the email lowercase.
2. **LEFT** to get the first initial.
3. **&** to join it with the last name and "@company.com".